

Lab 04 Template

1) Screenshot of pfSense pinging gateway (10 points)

```
Enter a host name or IP address: 42.49.30.254

PING 42.49.30.254 (42.49.30.254): 56 data bytes
64 bytes from 42.49.30.254: icmp_seq=0 ttl=63 time=0.238 ms
64 bytes from 42.49.30.254: icmp_seq=1 ttl=63 time=0.232 ms
64 bytes from 42.49.30.254: icmp_seq=2 ttl=63 time=0.225 ms

--- 42.49.30.254 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.225/0.232/0.238/0.005 ms

Press ENTER to continue.
```

**2) Discussion of pros/cons of port forwarding only vs virtual IPs AND port forwarding
(10 points)**

Port forwarding

Pros	Cons
It's easy to set up.	It has very limited flexibility because we can forward only 1 port per device, so it becomes difficult to manage as the number of devices increases.
It doesn't require additional IP addresses or configuration steps beyond routing specific traffic to internal devices.	It has a single point of failure.
It is easier to manage while we are using smaller networks	

Port forwarding with virtual IPs

Pros	Cons
It has greater scalability as the virtual IPs enable us to scale as it makes different IP addresses for different services, which makes the management easier as well.	More resources are required as they have more configuration and management to be looked for.
It has better security options as while using virtual IPs it lets us isolate and manage different services with more firewall rules	It has increased complexity as handling multiple IPs can lead to configuration errors, if they are not looked upon properly.
It allows to host multiple services using the same port numbers	

3) Screenshot of successful ssh connection from your ns1 to your desktop (20 points)

```
cpre230@ns1:~$ ssh pulkit@42.49.30.201
pulkit@42.49.30.201's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

43 updates can be applied immediately.
18 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

pulkit@desktop:~$ uname -a
Linux desktop 5.15.0-25-generic #25-Ubuntu SMP Wed Mar 30 15:54:22 UTC 2022 x86_64 x86_64 x86_64 GNU
/Linux
pulkit@desktop:~$ _
```

4) Screenshots (2) of Virtual IP & NAT Forwarding config pages (20 points)

Firewall / Virtual IPs

Virtual IP Address	Interface	Type	Description	Actions
42.49.30.201/24	WAN	IP Alias	desktop1	 
42.49.30.199/24	WAN	IP Alias	ns2	 
42.49.30.202/24	WAN	IP Alias	www	 
42.49.30.204/24	WAN	IP Alias	mail	 
42.49.30.205/24	WAN	IP Alias	ldap	 
42.49.30.206/24	WAN	IP Alias	www2	 
42.49.30.207/24	WAN	IP Alias	ws	 
42.49.30.208/24	WAN	IP Alias	splunk	 

+ Add

Firewall / NAT / Port Forward

Port Forward 1:1 Outbound NPt

Rules

	Interface	Protocol	Source Address	Source Ports	Dest. Address	Dest. Ports	NAT IP	NAT Ports	Description	Actions
☒	WAN	TCP	*	*	42.49.30.201	22 (SSH)	192.168.1.201	22 (SSH)	desktop - Nagios ssh check	  

Desktop - Nagios check 

↑ Add

↓ Add

 Delete

 Save

 Separator

5) Screenshot of successful Nagios check of ssh to desktop
(20 points)

Student 11 (student11)		
Host	Services	Actions
desktop1.student11.230.com	desktop1 ssh	
ldap.student11.230.com	ldap Bezos ldap Gates ldap Jobs ldap Musk ldap Zuckerberg ldap service	
mail.student11.230.com	dns MX mail pop secure imap secure imap login secure smtp	
ns1.student11.230.com	dns mail dns ns1 dns www dns www2 reverse dns mail reverse dns ns1 reverse dns www reverse dns www2	
ns2.student11.230.com	dns desktop1 dns ldap dns ws reverse dns desktop1 reverse dns ldap reverse dns ws	
www.student11.230.com	www http www https www login	
www2.student11.230.com	www2 django www2 ssh	

6) StudentXX.230.com diagram with only machines and services currently running
(20 points)

